

01:XXX:XXX - Homework n

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Contents

|       |                          |   |
|-------|--------------------------|---|
| 0.1   | Lecture 1: 9/1 . . . . . | 2 |
| 0.1.1 | Vector Norms . . . . .   | 2 |
| 0.2   | Lecture 2: 9/3 . . . . . | 4 |

## 0.1 Lecture 1: 9/1

### 0.1.1 Vector Norms

**Definition** (Vector Norm). A vector norm on  $R^d$  is a function  $\|\cdot\| : R^d \rightarrow R$  that satisfies the following properties:

1.  $\|x\| \geq 0$
2.  $\|ax\| = |a|\|x\|$  for all  $a \in R$
3.  $\|x + y\| \leq \|x\| + \|y\|$  (Triangle Inequality)

**Example** (p-norm). For  $p \geq 1$ , the p-norm is defined as

$$\|x\|_p = \left( \sum_{i=1}^d |x_i|^p \right)^{1/p}$$

The p-norm is a vector norm for  $p \geq 1$ .

**Theorem 1.** All p-norms are norms for  $p \geq 1$ .

*Proof.* We need to verify that the three properties of a vector norm hold for the p-norm.

1. Non-negativity: For any  $x \in R^d$ , we have  $|x_i| \geq 0$  for all  $i$ , so  $|x_i|^p \geq 0$  for all  $i$ . Therefore,

$$\|x\|_p = \left( \sum_{i=1}^d |x_i|^p \right)^{1/p} \geq 0.$$

Moreover,  $\|x\|_p = 0$  if and only if  $|x_i|^p = 0$  for all  $i$ , which is equivalent to  $x_i = 0$  for all  $i$ , i.e.,  $x = 0$ .

2. Homogeneity: For any  $a \in R$  and  $x \in R^d$ , we have

$$\|ax\|_p = \left( \sum_{i=1}^d |ax_i|^p \right)^{1/p} = \left( \sum_{i=1}^d |a|^p |x_i|^p \right)^{1/p} = |a| \left( \sum_{i=1}^d |x_i|^p \right)^{1/p} = |a| \|x\|_p.$$

3. Triangle Inequality: This is a consequence of Minkowski's inequality. For any  $x, y \in R^d$ , we have

$$\|x + y\|_p = \left( \sum_{i=1}^d |x_i + y_i|^p \right)^{1/p} \leq \left( \sum_{i=1}^d (|x_i| + |y_i|)^p \right)^{1/p}.$$

By Minkowski's inequality,

$$\left( \sum_{i=1}^d (|x_i| + |y_i|)^p \right)^{1/p} \leq \left( \sum_{i=1}^d |x_i|^p \right)^{1/p} + \left( \sum_{i=1}^d |y_i|^p \right)^{1/p}.$$

Therefore, we have

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

□

**Remark** (Weighted 2-norms). A weighted 2-norm is a vector norm of the form

$$\|x\|_W = \left( \sum_{i=1}^d w_i |x_i|^2 \right)^{1/2}$$

where  $w_i > 0$  for all  $i$ .

**Remark** ( $l_0$ -norm, sparsity norm). The  $l_0$ -norm is a "norm" that counts the number of non-zero entries in a vector. It is not a true norm because it does not satisfy the homogeneity property.

**Definition** (Holder's Inequality). For any  $p, q > 1$  such that  $\frac{1}{p} + \frac{1}{q} = 1$ , and for any vectors  $x, y \in R^d$ , we have

$$\sum_{i=1}^d |x_i y_i| \leq \|x\|_p \|y\|_q.$$

Cauchy-Schwarz inequality is a special case of Holder's inequality when  $p = q = 2$ .

**Definition** (Norm Equivalence). Two norms  $\|\cdot\|_a$  and  $\|\cdot\|_b$  on  $R^d$  are said to be equivalent if there exist positive constants  $c_1, c_2 > 0$  such that for all  $x \in R^d$ ,

$$c_1 \|x\|_a \leq \|x\|_b \leq c_2 \|x\|_a.$$

**Example.**  $\|x\|_1$  and  $\|x\|_2$  are equivalent norms on  $R^d$ . In fact, we have

$$\|x\|_1 \leq \|x\|_2 \leq \sqrt{d} \|x\|_1.$$

**Definition** (Matrix Norm). A matrix norm is a function  $\|\cdot\| : R^{m \times n} \rightarrow R$  that satisfies the following properties:

1.  $\|A\| \geq 0$  for all  $A \in R^{m \times n}$ , and  $\|A\| = 0$  if and only if  $A = 0$ .
2.  $\|\alpha A\| = |\alpha| \|A\|$  for all  $\alpha \in R$  and  $A \in R^{m \times n}$ .
3.  $\|A + B\| \leq \|A\| + \|B\|$  for all  $A, B \in R^{m \times n}$  (Triangle Inequality).
4.  $\|AB\| \leq \|A\| \|B\|$  for all  $A \in R^{m \times n}$  and  $B \in R^{n \times p}$  (Submultiplicativity).

**Example** (Frobenius Norm). The Frobenius norm of a matrix  $A \in R^{m \times n}$  is defined as

$$\|A\|_F = \left( \sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 \right)^{1/2}.$$

**Remark.**

$$\|A\|_F^2 = \text{Trace}(A^T A) = \sum_{i=1}^r \lambda_i A^T A$$

**Theorem 2** (Spectral Theorem). *If  $A \in R^{n \times n}$  is a symmetric positive semi-definite matrix, then there exists an orthogonal matrix  $Q \in R^{n \times n}$  and a positive diagonal matrix  $\Lambda \in R^{n \times n}$  such that*

$$A = Q\Lambda Q^T.$$

**Definition** (Spectral Norm). The spectral norm of a matrix  $A \in R^{m \times n}$  is defined as

$$\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}.$$

## 0.2 Lecture 2: 9/3

**Definition** (Moore-Penrose Pseudoinverse). For a matrix  $A \in R^{m \times n}$  and full rank, the Moore-Penrose pseudoinverse  $A^+ \in R^{n \times m}$  is the unique matrix that satisfies the following four properties:

1.  $AA^+ = I_n + 0_m$
2.  $A^+A = I_n$
3.  $(AA^+)^T = AA^+$
4.  $(A^+A)^T = A^+A$

Thus  $A^+$  is a generalization of the inverse matrix for non-square or singular matrices.

$$A^+ = (A^T A)^{-1} A^T \quad \text{if } A \text{ has full column rank}$$

And note that  $A^+ = V\Sigma^+U^T$  if  $A = U\Sigma V^T$  is the SVD of  $A$ .

**Theorem 3** (Eckart-Young-Mirsky Theorem). *Let  $A \in R^{m \times n}$  be a matrix of rank  $r$  with singular value decomposition  $A = U\Sigma V^T$ . For any integer  $k$  such that  $1 \leq k < r$ , let  $\Sigma_k$  be the diagonal matrix obtained by replacing all but the largest  $k$  singular values of  $\Sigma$  with zeros. Then the matrix*

$$A_k = U\Sigma_k V^T$$

*is the best rank- $k$  approximation to  $A$  in the sense that*

$$\|A - A_k\|_F = \min_{\text{rank}(B) \leq k} \|A - B\|_F.$$

$$A_s = \operatorname{argmin}_{\text{rank}(B) \leq k} \|A - B\|_2$$